Develop a website for PolyPhy

CROSS GSoC Contributor Proposal

CENTER FOR RESEARCH IN OPEN SOURCE SOFTWARE

Google Summer of Code

Asavari Ambavane

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Introduction

Abstract

Polyphorm is an interactive tool to analyze intergalactic gas and dark matter filaments (together known as 'Cosmic web') using the Monte Carlo Physarum Machine (MCPM) algorithm inspired by the foraging behavior of Physarum polycephalum 'slime mold'.

Problem

Through this project, we aim to create a website dedicated to this technology. To begin with, we will be incorporating information, guides, purposes, documentation, and much more about all the features that the PolyPhy technology offers us on this website. Interactive interfacing with appropriate linking is also an important part of the website. This project will serve as a platform for people to know about the software with all its details.

Background

Currently what we have is a rich repository of the code and documentation of the project. We have to port all of that to a web interface. Polyphorm is an agent-based system for reconstructing and visualizing optimal transport networks defined over sparse data. It uses a richer 3D scalar field representation of the reconstructed network. Whereas, PolyPhy will be a Python-based redesigned version of Polyphorm, currently at the beginning of its development cycle. PolyPhy will be a multi-platform toolkit meant for a wide audience across different disciplines: astronomers, neuroscientists, data scientists, and even artists and designers. Our goal through this project is to create an integrated hub for passionate contributors to find out about everything PolyPhy.

Solutions

Our job is to create a dynamic website housing the following things. I will be explaining how I plan on implementing all this in more detail as we move on further.

- <u>About PolyPhy</u>
  - A brief overview
  - It's origin
  - Algorithm on which it is based
  - Features
  - The authors
- <u>How to go about using it</u>
  - Installation Instructions
  - How to give input and get outputs
  - Controls and functions
  - Various types of visualizations
- <u>Reference Materials</u>
  - Video tutorials
  - Research paper links
  - Links to google colab notebooks as a reference for people.
  - Github Repository
- <u>Community Portal</u>
  - Blogs
  - Slack
  - Ask doubts - link to stack overflow or GitHub forums
  - Tweets about PolyPhy
- <u>Contribution guide</u>
  - Github issues
  - About making PRs

Biographical Information

Educational background

I am currently a third-year undergraduate majoring in Electronics Engineering at VJTI, Mumbai. I completed my high school education at Smt. Sulochanadevi Singhania School scored an aggregate of 97.5% in my examinations. I am hardworking when it comes to academics and hold a CGPA (Cumulative Grade Point Average) of 9.2 on a scale of 1 to 10. In my college, I actively participate in technical as well as non-technical activities. I hold the position of Chief Management Officer at Pratibimb, the cultural fest of our college. Other than that, I am a part of GDSC, The Google Developer Students Club google facilitated technologies such as Cloud Computing and Flutter.

Technical Interests

I am passionate about the field of software development niching down to web development and am proficient in technologies and languages such as React JS, JavaScript, NodeJS, NextJS, etc. I have consistently worked on projects entailing the same over the course of my college term. But, I am relatively new to open source and therefore I am looking forward to growing and developing these skills more and more by contributing to this platform.

Some of my projects are -

1. Pratibimb VJTI
   Designed and developed the official website of Pratibimb VJTI, the cultural extravaganza of VJTI Mumbai. It showcases the organization's glamour through a rich variety of activities, events, and public relations.

2. EPINFO
   Collaborated on a project under the health-tech domain to create an epidemiological infographics dashboard as a part of a 2-day DevFest. It aims to simplify epidemiological studies while taking into account the anonymity of users.

3. Place of Thoughts
   Established an interface similar to Medium. This web app provides a platform for users to publish their own pieces. Users can leave comments on fellow writers' posts and even save articles for reading later.

Prior Experience with open source

1. [User Story - EOS]
   - ➢ Made pages of the app mobile responsive
   - ➢ Enhanced app UI such that it looks good on small size devices too.
   - ➢ PR merged

2. [Recess]
   - ➢ Opened Issue
   - ➢ Provided a UI for chats

Communication

Working Hours - I am flexible with my timings and can work dedicatedly for hours together. I would like to work during the daytime as I find myself more productive during the early hours. Also, I will make sure that GSoC will be my only major commitment this summer and that I give it my full attention.

Communication preferences - I am most comfortable with the google meet or zoom platform but can adjust to whatever suits the mentors.

In case of work delay due to personal or other reasons, I assure you that I will work longer and efficiently to complete the backlog in the available time. I plan on writing weekly progress blogs on my medium publication to establish a systematic solution to prevent missing out on a week's work. I would be sharing the same with my mentors while having active communication with them.

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Github - https://github.com/AsavariA

Twitter - https://mobile.twitter.com/asaworryy

More About Me

I am a trained Bharatnatyam dancer though now I dance as just a hobby. I also enjoy listening to music and reading books in my free time. Soft pop and indie are the genres I love and groove to. When it comes to books, I generally prefer a book with brilliant adventure or suspense and themes of cryptography and conspiracy theories. Dan Brown is one of my favorite authors closely followed by Rick Riordian. Also, I have never lived far from my family and that's a major reason why I have bittersweet feelings about leaving home for my post-graduation. I have a really creative twin brother who has a design firm of his own even at this age. We manage this firm together - I lead the technical projects while he designs products for clients.

Resume: Asavari Ambavane resume.pdf

Project Objectives and Expected Results

The primary objective of this project is to help disseminate PolyPhorm to the public, more importantly, the technology enthusiasts of the world.

As I have mentioned before, open source is still a pretty novel term to me and this is my first time participating in GSoC. Thus, I want to explore it as much as possible so that I am able to grow and develop my skillset in this productive atmosphere. Gaining a sense of independence is also one of my top desires for this position. I've always worked on tight-knit teams, which has been wonderful, but I am ready for a chance to prove what I can do on my own.

I am looking forward to this project also because it is an exciting opportunity for my career. Most importantly, through participating in GSoC and in turn PolyPhorm, I want to give back to the huge community of developers out there which has been one of my long-term goals.

This website will serve the purpose of increasing awareness regarding the study of cosmological datasets facilitated by this novel tool. We can all agree that a product such as this is a huge boon to the community but advertising it, branding it, and placing it in the public eye can help it gather the amount of reach it deserves. Thus, through this website, I hope that the community discovers and recognizes PolyPhy as the useful tool that it is.

Why UC Cross?

This project offered by UC Cross has my two favorite things meshed perfectly around each other. Web and Open Source. UC Cross is definitely one of the best organizations and when I found out that it was open-source, I was thrilled just by the idea to get to be a part of it. I think it has everything I wanted in an ideal organization - be it values, scope, future, opportunities, and everything else. Thus this has always been a dream of mine and I look forward to working here!

Deliverables and Implementation Plan

After going through the repository for this project, I came up with a few ideas on how I think I would go about this project during GSoC. I mentioned all those ideas briefly above. In continuation, I would like to elaborate on those further. The website will have the following features.

1. **Landing Page**
   This will be the starting point of our website. It will have the following features.
   a. The landing page's headline is the very first thing visitors will see. For this reason, it is essential that it describes very clearly what a user will get from the page, in our case, PolyPhorm
   b. Brief abstract about the project - This will pique visitors' interest in knowing more about the package.
   c. Features of PolyPhy i.e. what does PolyPhy has to offer.
   d. General information about the Physarum polycephalum 'slime mold' and Monte Carlo Physarum machine - displayed with accordions.

2. **Installation Guide**
   An important part of any software website is an installation guide.
   a. Any kind of prerequisites will be mentioned here.
   b. Since it's a python package, steps for installation using popular package managers like `pip` and `conda` will be written on this page.
   c. Different OSs' have different ways to install, even that will be covered here.

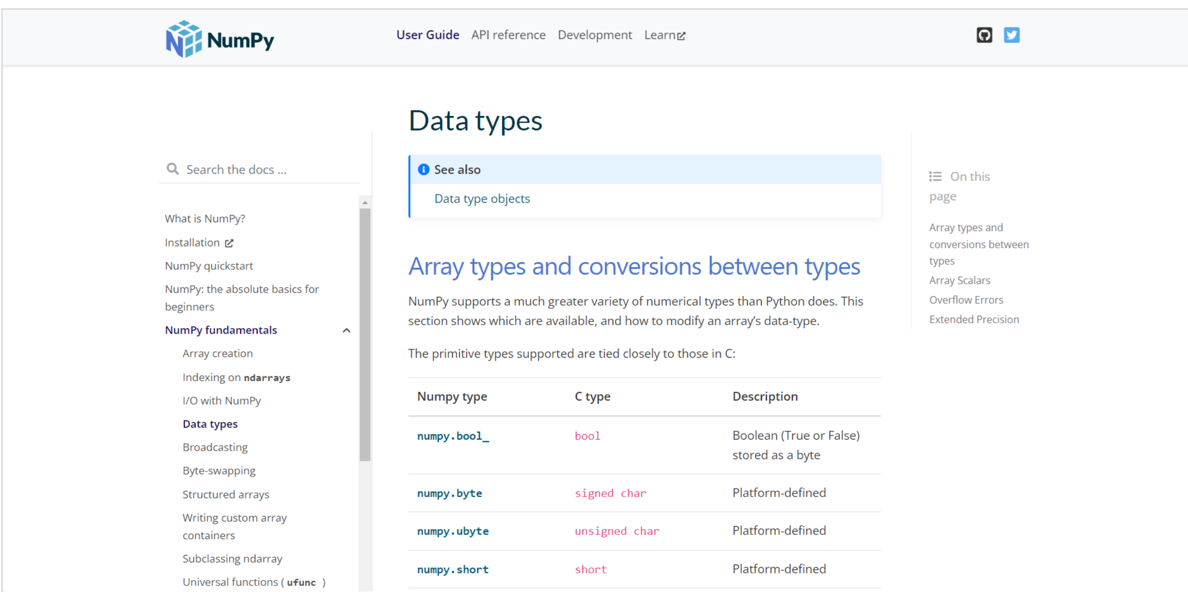
3. **Learn about PolyPhy**
   a. This section will be for those inquisitive minds who want to explore more reference articles, video tutorials, books, use-cases, etc.

b. This section may also include embedded code snippets of the different results we can achieve using the tool.

4. **Documentation**

   Here the idea is to generate the documentation automatically through some software (like Doxygen, Natural Docs, etc)

   a. Make proper documentation for PolyPhy and port it onto the website on a new page
   b. Documentation will be in .md files therefore, we need to provide support to view these files on the web.
   c. Rough prototype inspired from https://numpy.org/
   d. This will include input, output, visualization types, functions, controls, and similar things which will ease the process of writing code for the user.



5. **Contributions and open source guide**

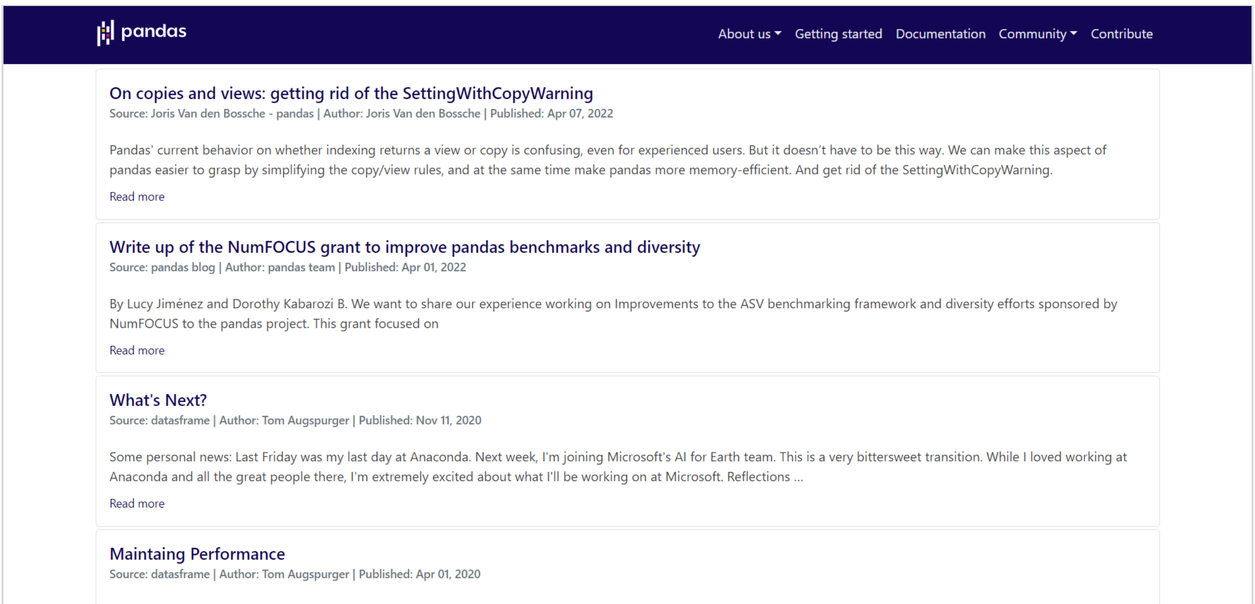
   What is an open-source software without its contributors?

   a. Contributors guide - how to open good PRs
   b. How to open issues
   c. The authors and original contributors of the project
   d. Donations if any

6. **Community Portal**

   This is a very fun and interesting part of any product website. Not only the authors but also the experienced users of this product write about it. How exciting is that! There are several ways to do this, some of the approaches are -

a. Write all blogs on a single platform and fetch them through APIs eg. Medium, Github, Dev To.
b. Get all blogs in .md files and display them on the PolyPhy website.
c. Store brief information (like title, subtitle, publishing date, tags) about all blogs in a simple database such as firebase firestore and fetch them onto the PolyPhy website. We can then link them to their respective platforms. (reference - [python pandas website blog](#))



7. **Mobile Responsiveness**
I will make sure that the website looks just as good on mobiles and devices of other sizes as it does on laptops/pc screens. This is a very crucial user experience feature one should never neglect.

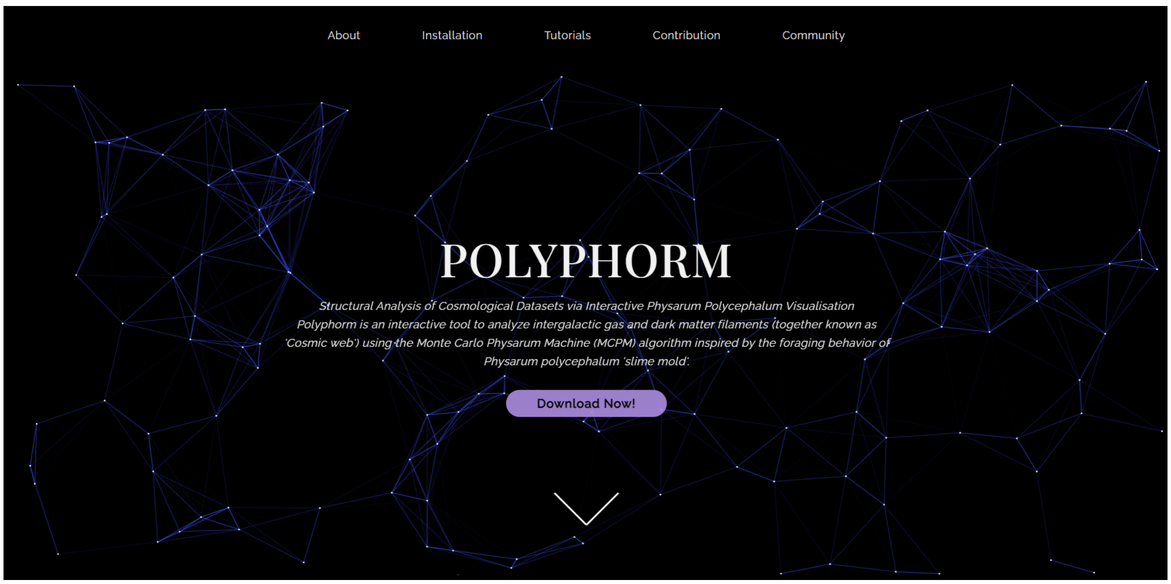
8. **Easy documentation and maintenance of the website**
While creating an open-source website, I would keep in mind the need for it to be maintained and extended from time to time. To achieve this, primarily I would create a readme in the Github repository of the website and include a guide of -
    a. File structure
    b. Javascript and stylesheet format
    c. How to clone and go about running and extending it, is basically a contributor's guide.

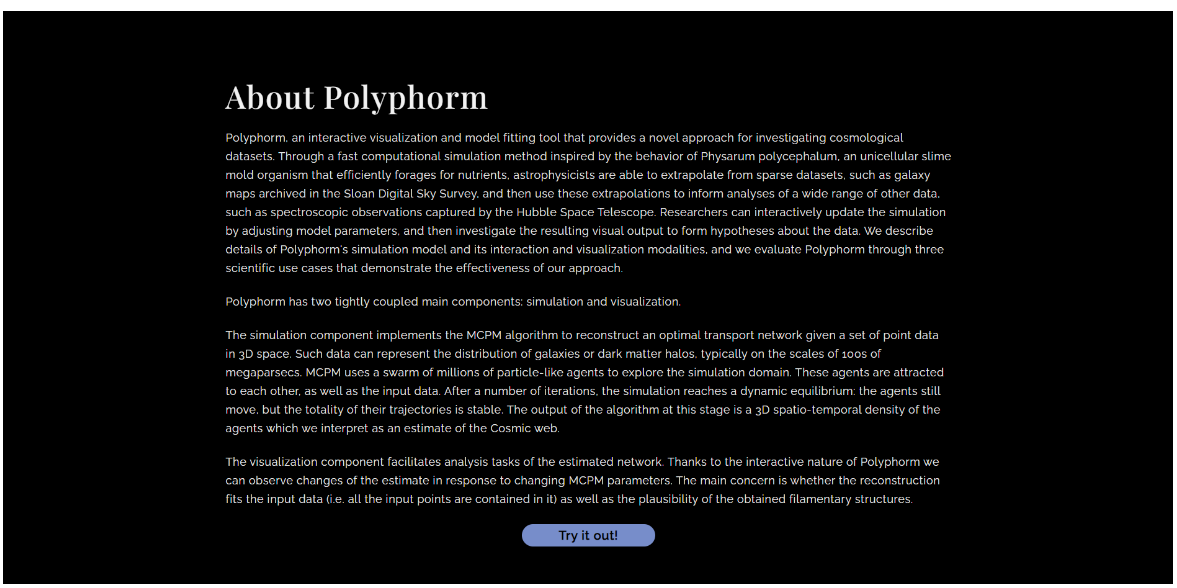
Code Affected

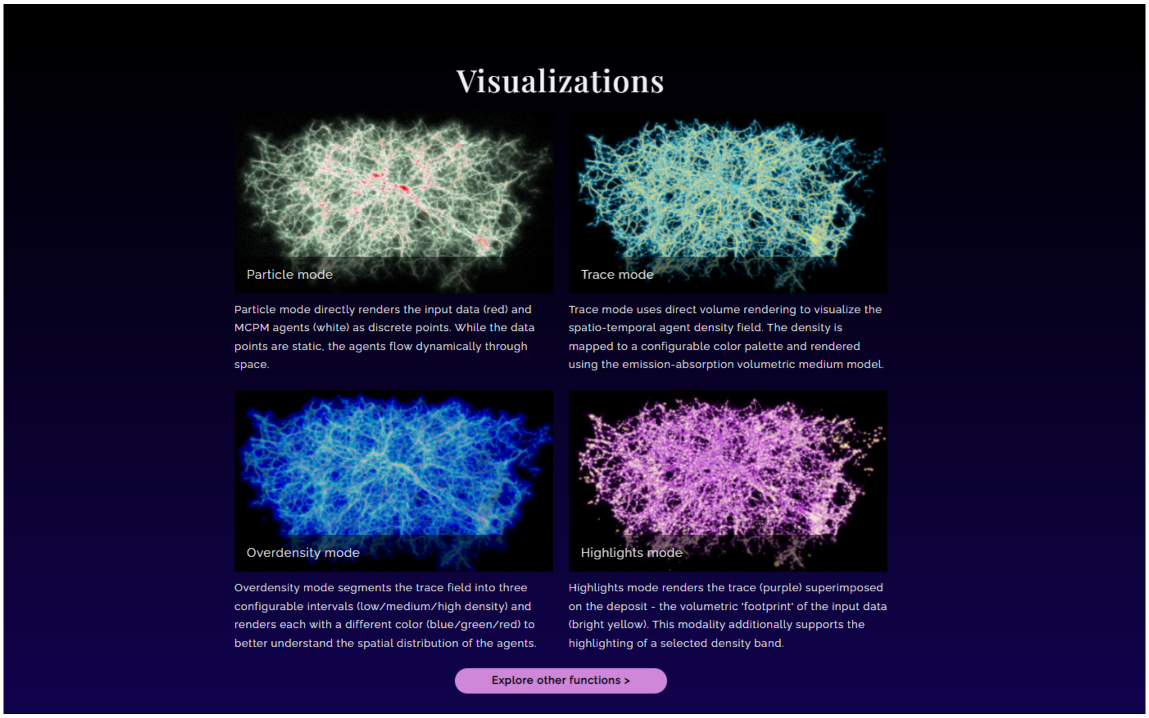
This project will start from scratch and hence no previous code will be affected.

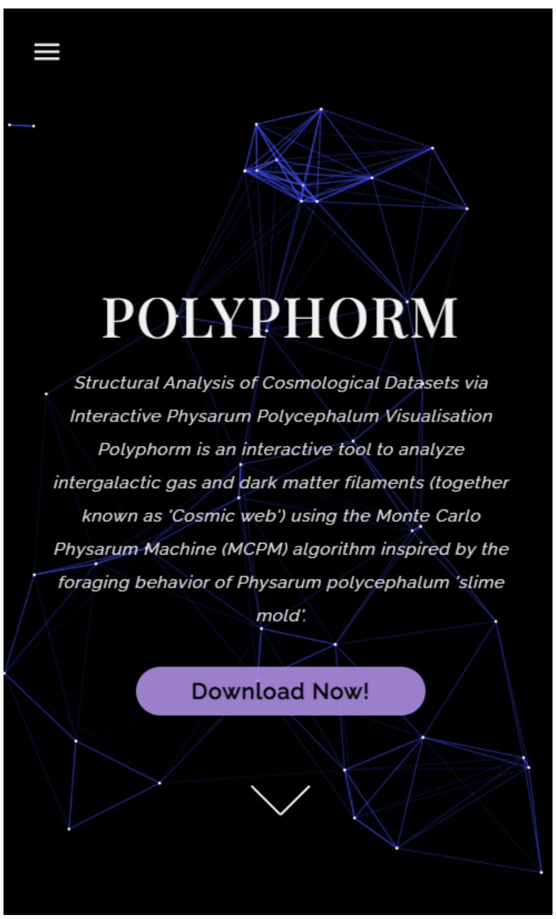
Related Pre Proposal Work

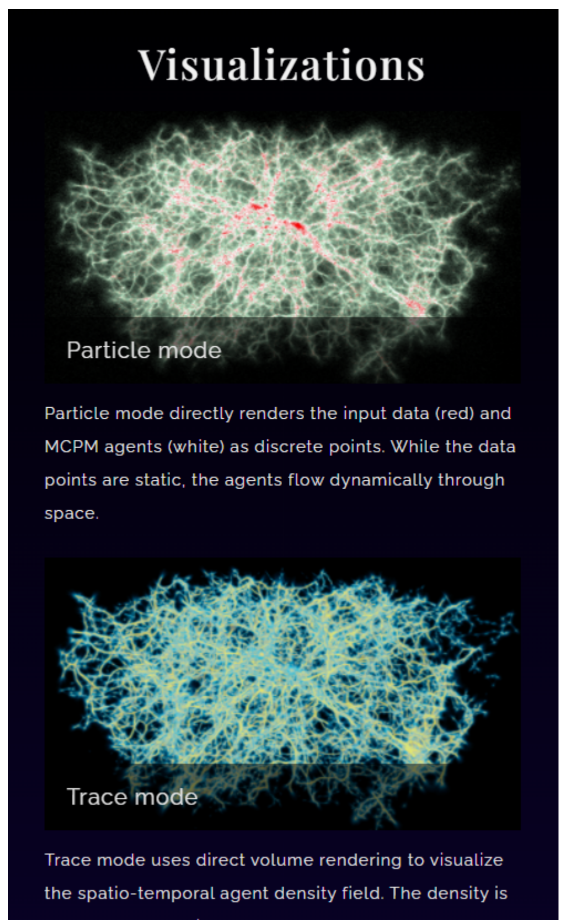
Here are some rough prototypes I made before diving into the proposal just for the mentor's reference on how I imagined the website to look like. This prototype is fully responsive plus has a beautiful UI.






Visualizations
Particle mode
Trace mode
Particle mode directly renders the input data (red) and MCPM agents (white) as discrete points. While the data points are static, the agents flow dynamically through space.
Trace mode uses direct volume rendering to visualize the spatio-temporal agent density field. The density is mapped to a configurable color palette and rendered using the emission-absorption volumetric medium model.
Overdensity mode
Highlights mode
Overdensity mode segments the trace field into three configurable intervals (low/medium/high density) and renders each with a different color (blue/green/red) to better understand the spatial distribution of the agents.
Highlights mode renders the trace (purple) superimposed on the deposit - the volumetric 'footprint' of the input data (bright yellow). This modality additionally supports the highlighting of a selected density band.
Explore other functions >


POLYPHORM
Structural Analysis of Cosmological Datasets via Interactive Physarum Polycephalum Visualisation
Polyphorm is an interactive tool to analyze intergalactic gas and dark matter filaments (together known as 'Cosmic web') using the Monte Carlo Physarum Machine (MCPM) algorithm inspired by the foraging behavior of Physarum polycephalum 'slime mold'.
Download Now!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Schedule of Deliverables

My weekly plan and estimates for working on the deliverables.

April 20 - May 20 (Pre Selection)

Week 1 - Find inspiration for website design.

Week 2 - Start Prototyping the website UI

Week 3 - End Semester Exam Break

Week 4 - Finish the UI and get it approved by the mentor.

May 20 - June 12 (Community Bonding Period)

Week 1 - Get to know the team and mentors and discuss the project with them.

Week 2 - Write and gather content for the website.

Week 3 - Write installation guides, collect tutorials and reference articles from the PolyPhy team

June 13 - July 25 (Phase I)

Week 4 - The coding Period starts! Start hands-on work with building the website. Build common components like responsive navbar and footer.

July 25 - September 12 (Phase II)

Week 5 & 6 - Start the landing page of the website.

Week 7 - Port the installation procedure on the website in .md format.

Week 8 - Similarly, work on displaying the authors and guide for contributors for opening PRs with this tool.

Week 9 - Making 2 weeks of work responsive.

Week 10 & 11 - Generate the extensive documentation by code and display it on the web. Add code snippets, examples, and results to all the functions PolyPhy offers.

Week 11 - Create a community section, wherein I will map over blogs. During this time, I will also work with the Twitter API to parse tweets with PolyPhorm in them and display them. Let's link 'ask a question' to GitHub issues or stack overflow, followed by 'Code of Conduct'

Week 12 - Making the whole website responsive

September 12 - November 21 (For Extended Timelines)

Finish up the last few things and final changes before the end of the program.

Post GSoC Plans

I plan to harvest the experience and knowledge I will gain from GSoC and move forward to making regular contributions to open source projects. I am fairly new to the concept.

I have been doing projects and uploading them publicly for a year now but wasn't aware of the big community behind many of the tools I have been using.

I would like to show people my work and ask them to join in on the mission whilst also contributing to other projects to gain experience and pass it on.

Also, I plan to help my juniors prepare for next year's GSoC, introduce them to open source, and help them get a head start on projects my club mates and I have worked on in GSoC.